Introduction to AI in Finance

BUS 696: GENERATIVE AI IN FINANCE

WEEK 1

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Today's Agenda

01

Course Overview

Understanding expectations and learning objectives for the semester

02

Historical Evolution

From human traders to algorithmic intelligence

03

Finance Fundamentals

Stocks, returns, and risk metrics explained

04

Python for Finance

Understanding and analyzing financial data

05

Prompt Engineering

Crafting effective instructions for AI systems

06

Hands-On Lab

Practical application of today's concepts

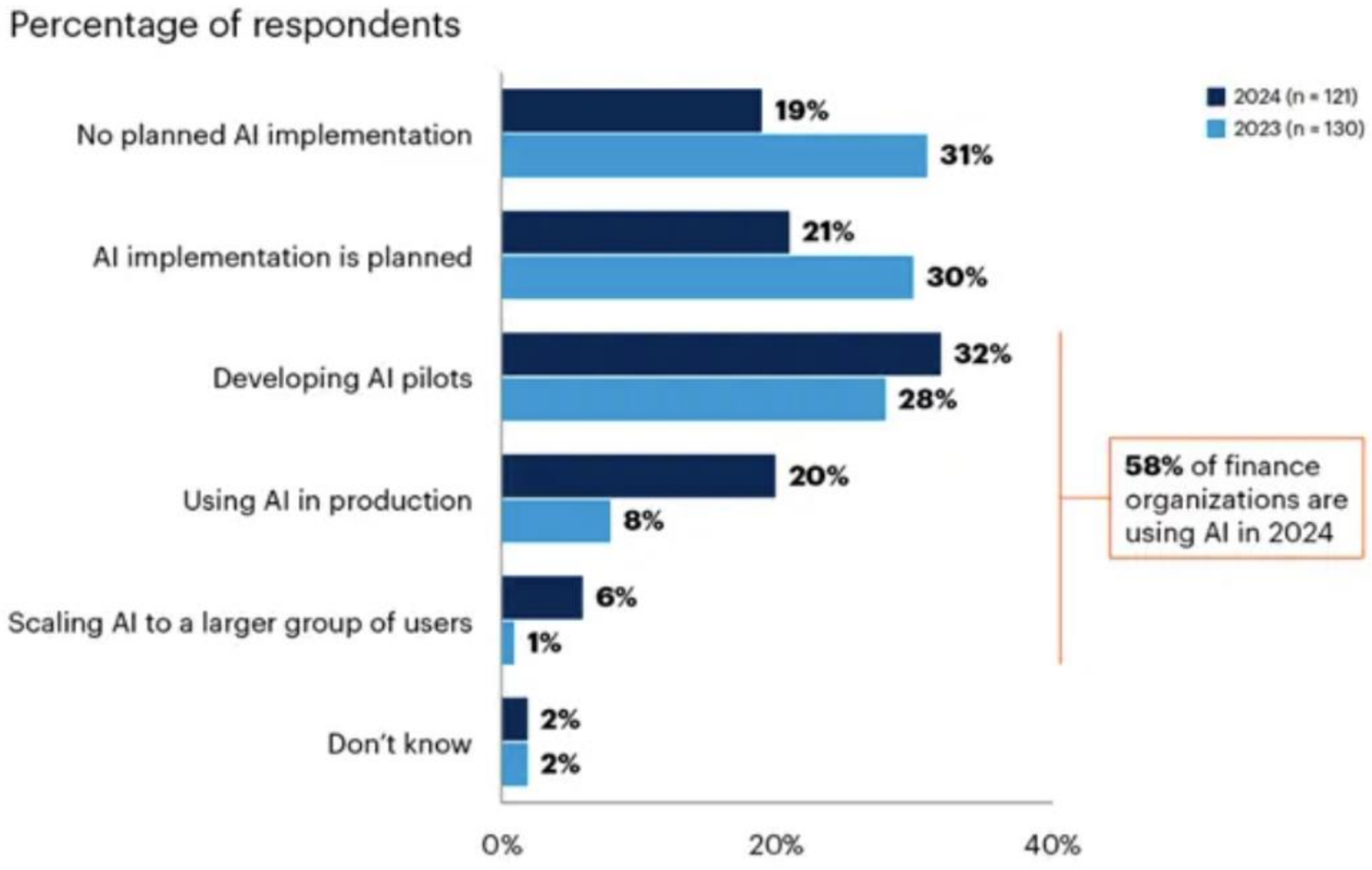
What This Course Is About

AI is fundamentally transforming every aspect of finance, from trading to risk management. This course takes a hands-on approach combining readings, discussions, and practical coding exercises.

No finance background required—we build from scratch. **No heavy coding**—we modify pre-built notebooks to understand concepts.

Our goal: develop the knowledge to critically evaluate how AI works in finance and understand its real-world applications.

Current AI Use in Finance



n varies; Finance leaders

Source: 2024 and 2023 Gartner AI in Finance Survey

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Why AI in Finance Now?

Three powerful forces are converging to make this the age of AI-driven finance:

Data Explosion

Markets generate terabytes of data daily. Alternative data sources—satellites, social media, web scraping—provide unprecedented insights into market behavior and economic trends.

Compute Power

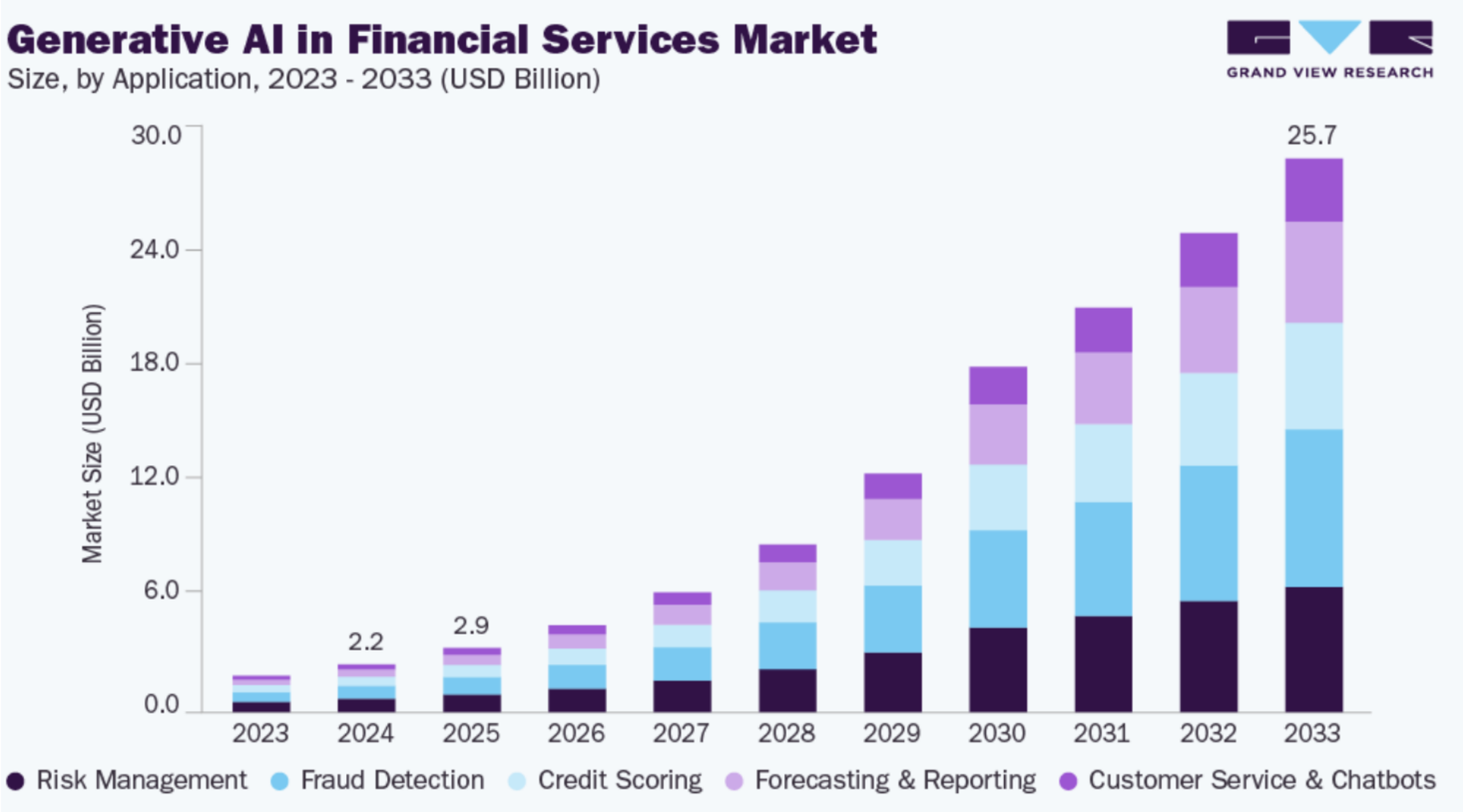
GPU technology makes training neural networks feasible at scale. Cloud computing has democratized access to powerful computational resources once available only to elite firms.

Algorithmic Breakthroughs

Deep learning, transformers, and large language models can identify patterns and relationships that humans would never detect, revolutionizing predictive capabilities.

Result: Machines now execute over 70% of all stock trades, fundamentally changing market dynamics.

What Are Firms Using AI For?



AI is Impacting Every Part of Finance: PE Deals

What a Wild Time to Be Alive

Pinned

Legendary
@Legendaryy

My clawdbot just asked me for an RTX 4090. Instead of buying it, I gave it a $2K trading wallet on Hyperliquid.

I said: If you want the GPU, earn it. It now trades crypto, stocks, and commodities 24/7.

It scans Twitter sentiment, tracks Trump posts, and decides trades on its own. Every trade gets logged. It scores what worked, drops what didn't.

🎯 **Trading Model**

```
SCAN → SIGNAL → RESEARCH → DECIDE → EXECUTE → MONITOR → CLOSE
  |       |        |        |        |        |        |
 4h     Price/   Perplexity Long/   Max 10%  30min   Journal
cron    News/   deep dive  Short/  3x lev   check   + Learn
        Trump              Wait    Stop-loss
```

🔄 **Self-Learning Loop**

```
TRADE → JOURNAL → EXTRACT LEARNING → APPLY
                        |
                  Salience Score
                        |
        ┌───────────────┼───────────────┐
        ▼               |               ▼
   < 0.2 ARCHIVE        |        > 0.8 PROMOTE
                        ▼        to TRADING_DOCS
              0.2–0.8 ACTIVE
              (in learnings.md)
```

Salience Changes:
- Trade won with learning: +0.1
- Trade lost despite learning: −0.1
- Time decay: −0.02/week

10:36 AM · Jan 24, 2026 · **1.7M** Views

https://medium.com/@luoyelittledream/building-an-ai-powered-automated-trading-system-from-scratch-making-clawdbot-openclaw-your-4294f0c05847

Stockbench: Which AI Model Performs Best?

STOCKBENCH

Abstract Overview Targets Leaderboard

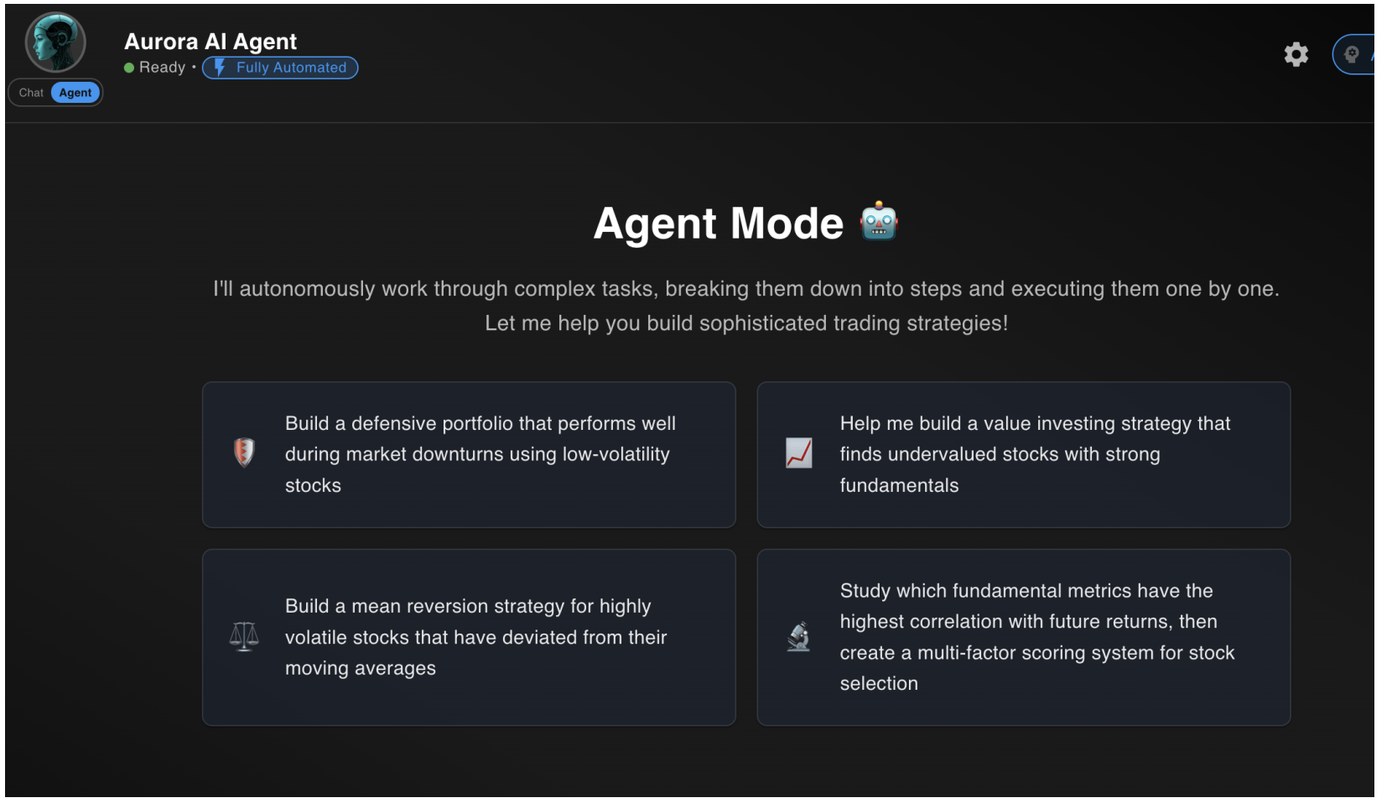
Leaderboard

Performance comparison of state-of-the-art models

RANK	MODEL	FINAL RETURN (%)	MAX DRAWDOWN (%)	SORTINO RATIO
1	**Kimi-K2**	+1.9	-11.8	0.0420
2	**Qwen3-235B-Ins**	+2.4	-11.2	0.0299
3	**GLM-4.5**	+2.3	-13.7	0.0295
4	Qwen3-235B-Think	+2.5	-14.9	0.0309
5	OpenAI-O3	+1.9	-13.2	0.0267
6	Qwen3-30B-Think	+2.1	-13.5	0.0255
7	Claude-4-Sonnet	+2.2	-14.2	0.0245
8	DeepSeek-V3.1	+1.1	-14.1	0.0210
9	GPT-5	+0.3	-13.1	0.0132

https://stockbench.github.io/

Aurura AI Agent



LLMs will perform poorly without sufficient context – so let's just give the models a lot of context!

https://nexustrade.io/agent

https://nexustrade.io/blog/too-many-idiots-are-using-openclaw-to-trade-heres-how-to-trade-with-ai-the-right-way-20260203

A Brief History

The evolution from human-driven markets to AI-powered finance:

1. **1602**
   First stock exchange established in Amsterdam
2. **1967**
   First handheld calculators appear on trading floors
3. **1971**
   NASDAQ launches as first electronic stock exchange
4. **1983**
   Bloomberg Terminal revolutionizes financial data access
5. **2000s**
   Rise of quantitative hedge funds like Renaissance and DE Shaw
6. **2010s**
   High-frequency trading dominates; robo-advisors emerge
7. **2020s**
   LLMs enter finance; AI agents begin autonomous trading

What is a Stock?

A stock represents an ownership share in a company. Its value is the present discounted value of future cash flows of the company

Example: If Apple has 16 billion shares outstanding and you own 100 shares, you own 0.000000625% of Apple.

The **stock price** reflects what buyers are willing to pay at any given moment. Prices fluctuate every second based on supply and demand.

When you buy a stock, you're betting the company will become more valuable over time.

> **Real Example:** Apple stock price today ≈ $180. Buying 10 shares = $1,800 invested.

What is a Return?

A **return** measures how much your investment gained or lost, expressed as a percentage of the original investment.

$$Return = \frac{Price_{today} - Price_{yesterday}}{Price_{yesterday}}$$

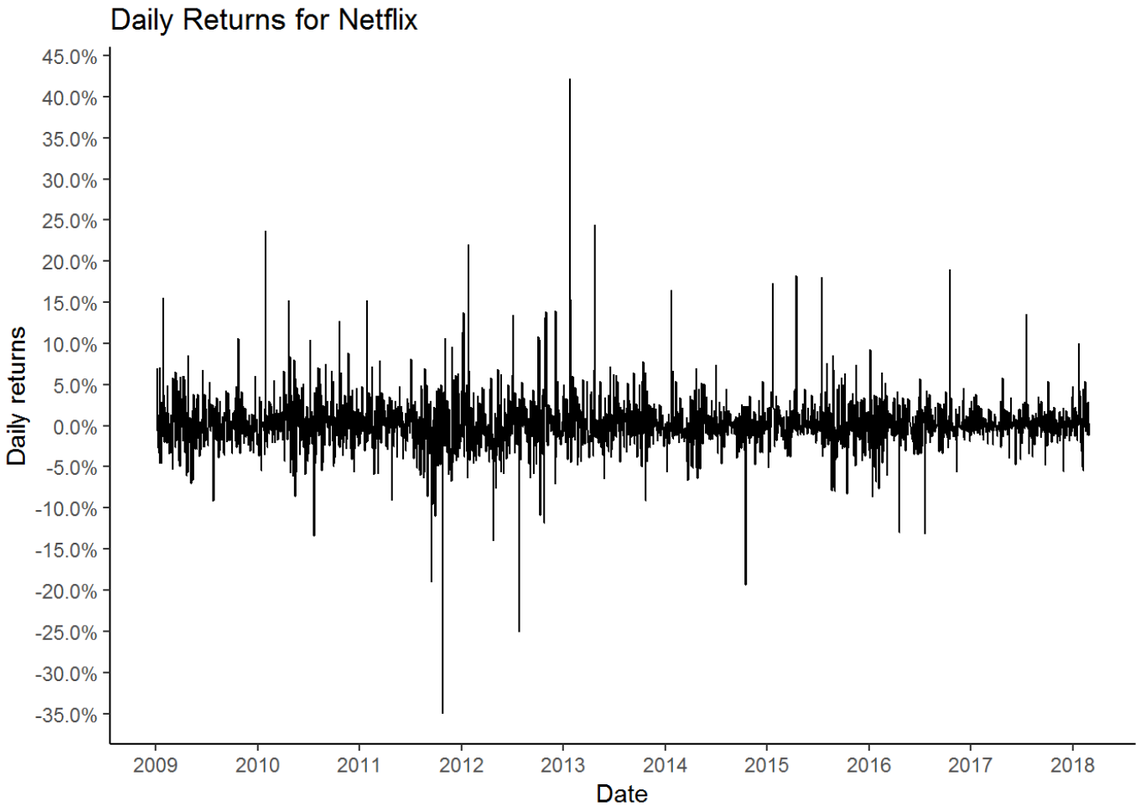
Positive Return Example

- Yesterday: Apple = $180
- Today: Apple = $183
- Return = ($183 - $180) / $180 = 0.0167
- **Result: +1.67% gain**

Negative Return Example

- Yesterday: Apple = $180
- Today: Apple = $176
- Return = ($176 - $180) / $180 = -0.0222
- **Result: -2.22% loss**

Daily Returns in Practice



Daily returns behave like a seismograph—mostly small tremors with occasional earthquakes.

Typical patterns:

- Most days: small moves between -1% and +1%
- Occasionally: big moves triggered by earnings, Fed decisions, or market crises
- Returns fluctuate around zero—no consistent direction

Key insight: Returns are inherently noisy and extremely difficult to predict on a day-to-day basis. This randomness is what makes trading challenging.

What is Risk?

In finance, **risk equals volatility**—specifically, how much returns bounce around from day to day.

We measure risk using the **standard deviation** of returns:

Low Volatility

~1% daily standard deviation

Stable, predictable performance. Think utility companies—boring but reliable.

High Volatility

~4% daily standard deviation

Wild swings in value. Think biotech startups—exciting but nerve-wracking.

The fundamental tradeoff: Higher expected returns require taking on more risk. There's no free lunch in investing.

Understanding this relationship is crucial for making informed investment decisions and managing portfolios effectively.

The Sharpe Ratio

How do we compare investments with different risk levels? Enter the Sharpe Ratio.

$$Sharpe\ Ratio = \frac{Return - Risk\text{-}free\ rate}{Volatility}$$

In plain English: "How much extra return do I get per unit of risk I'm taking?"

Investment A

12% return, 20% volatility

Sharpe = 0.6

Investment B

8% return, 10% volatility

Sharpe = 0.8

Investment B is better! You get more return per unit of risk, with less volatility to endure.

25%

Poor

Sharpe < 0.5

50%

Acceptable

Sharpe 0.5-1.0

75%

Good

Sharpe 1.0-2.0

95%

Excellent

Sharpe > 2.0 (or suspicious!)

What is a Moving Average?

A **moving average** smooths out daily price noise to reveal underlying trends.

The **20-day moving average** calculates the average of the last 20 closing prices.

Example calculation:

- Day 1-20 prices: $180, $182, $179, ... (20 values)
- 20-day MA = sum of those 20 prices ÷ 20 = $181.50

Each day, the window "moves" forward—drop the oldest price, add the newest.

Price Above MA

Uptrend, bullish signal indicating positive momentum

Price Below MA

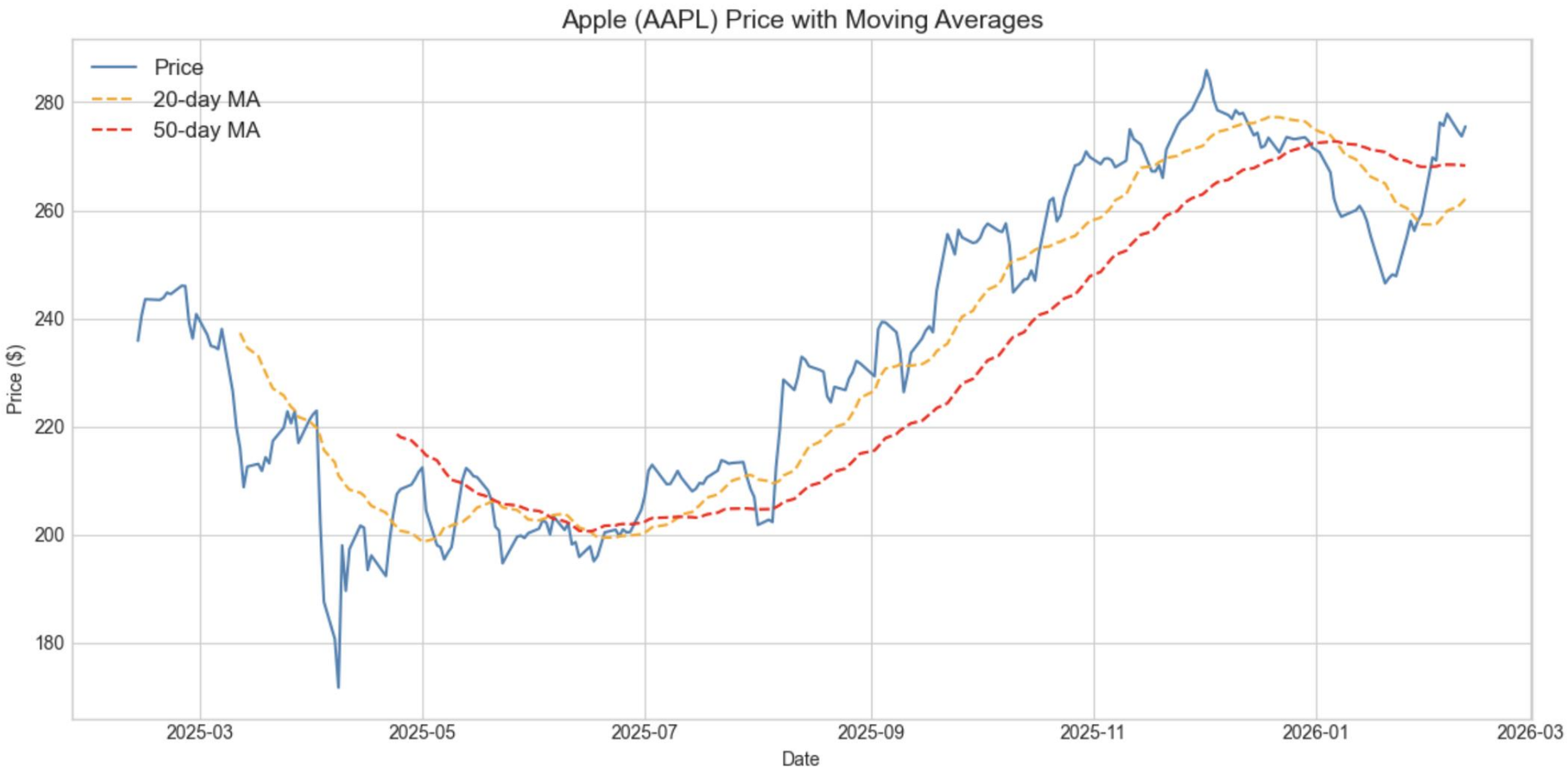
Downtrend, bearish signal suggesting weakness

Price Crosses MA

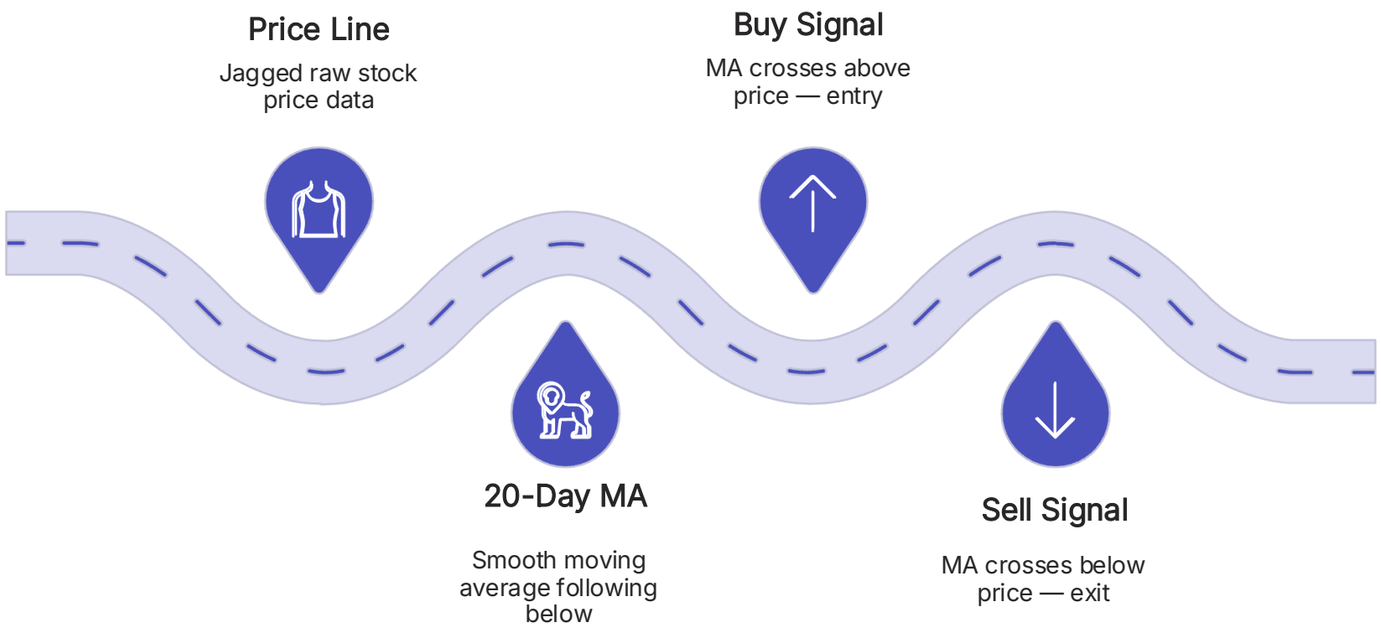
Potential trend reversal requiring attention

Common periods: 20-day (short-term), 50-day (medium), 200-day (long-term)

Moving Average



Moving Average Visualization



The chart illustrates how moving averages filter out market noise to reveal trends.

Key observations:

- Raw price data appears jagged and erratic
- Moving average creates a smooth line following price action
- MA lags behind price (based on historical data)
- Price crossing above MA = potential buy signal
- Price crossing below MA = potential sell signal

This is a simple example of a **trading signal**. AI can identify far more complex patterns across multiple indicators simultaneously.

Getting Financial Data with Python

We use **yfinance**, a free Python library that accesses Yahoo Finance data to download historical stock information.

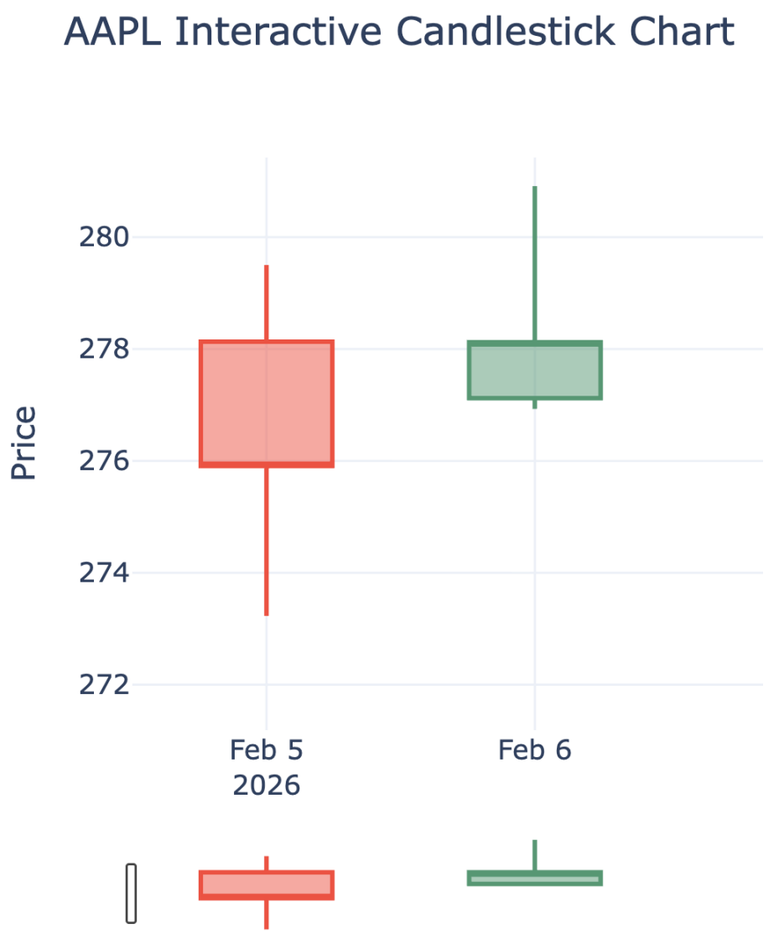
```
import yfinance as yf

# Download Apple's last year of data
apple = yf.download('AAPL', period='1y')
```

What you receive:

2025-02-10	179.50	181.20	178.90	180.25	45M

- **Open:** Price at market open (9:30 AM ET)
- **High/Low:** Highest and lowest prices during the trading day
- **Close:** Price at market close (4:00 PM ET)—this is the primary reference price
- **Volume:** Total number of shares traded

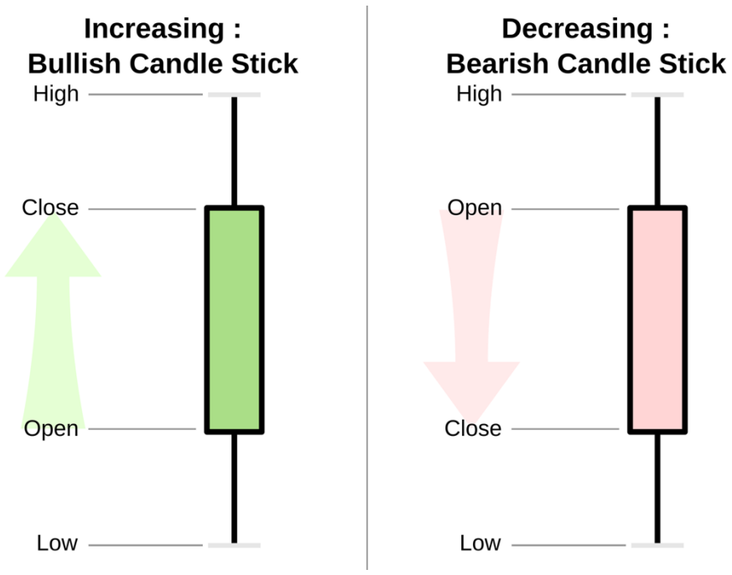
Candlesticks



Candlesticks graph out the open, close and low and high of a stock over a period

If the closes higher than it opened -> bullish and green

If the closes lower than it opened -> bearish and red



Calculating Returns in Python

```
# Calculate daily returns
apple['Return'] = apple['Close'].pct_change()

# Get summary statistics
apple['Return'].mean()  # Average daily return
apple['Return'].std()   # Volatility (std dev)

# Annualize the metrics
annual_return = apple['Return'].mean() * 252
annual_vol = apple['Return'].std() * (252 **
0.5)
```

The `pct_change()` function automatically computes the percentage change formula:

(today - yesterday) / yesterday

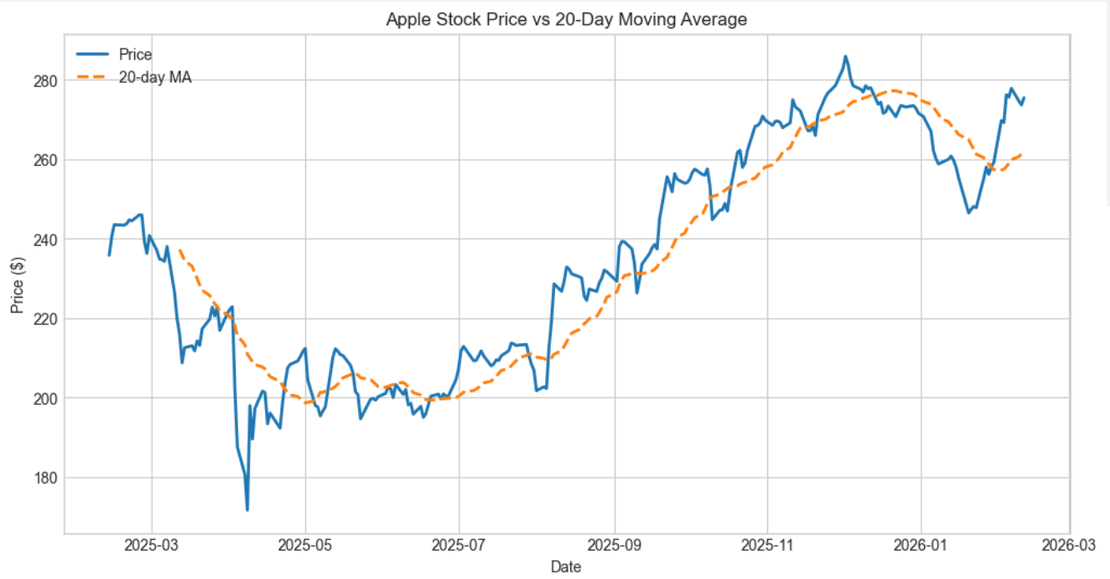
Annualization factors:

- Annual return ≈ daily return × 252 trading days
- Annual volatility ≈ daily volatility × √252

These conversions allow you to compare investment performance on a consistent annual basis.

Calculating Moving Averages in Python

```
# Calculate 20-day moving average
apple['MA20'] = apple['Close'].rolling(window=20).mean()

# Plot price versus moving average
import matplotlib.pyplot as plt

plt.figure(figsize=(12, 6))
plt.plot(apple['Close'], label='Price', linewidth=2)
plt.plot(apple['MA20'], label='20-day MA', linewidth=2, linestyle='--')
plt.legend()
plt.title('Apple Stock Price vs 20-Day Moving Average')
plt.xlabel('Date')
plt.ylabel('Price ($)')
plt.show()
```



How it works:

- `rolling(window=20)` = "look at the last 20 rows of data"
- `.mean()` = "calculate their average"
- First 19 days will show NaN (not enough data yet)

Interpreting Stock Statistics

When analyzing stock performance, understanding key metrics is essential. Here's how to read common statistics:

Daily Return: 0.05%

0.05% × 252 trading days ≈ **12.6% annual return**

This is solid performance, above historical market averages.

Daily Std Dev: 1.8%

1.8% × √252 ≈ **28.5% annual volatility**

Fairly risky—expect significant price swings throughout the year.

Sharpe Ratio: 0.44

12.6% return ÷ 28.5% volatility = **0.44**

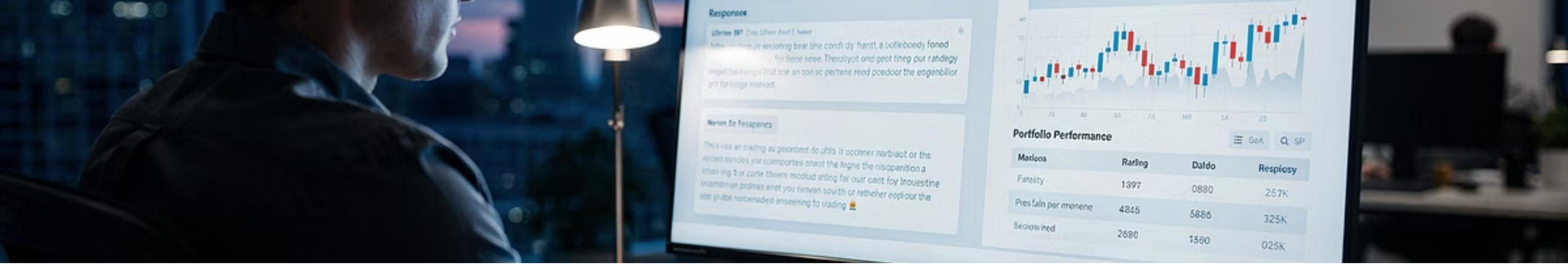
Mediocre risk-adjusted performance. Not enough return for the risk taken.

Total Return: 15%

$1,000 invested → **$1,150**

Absolute gain of $150 over the period.

Context matters: Always compare to benchmarks like the S&P 500 (historically ~10% annual return, ~15% annual volatility).



What is Prompt Engineering?

A **prompt** is the instruction you give to an AI system. The quality of the AI's output depends heavily on the quality of your input.

Bad Prompt

"Tell me about Apple stock"

Result: Generic, unfocused response with little actionable insight

Good Prompt

"Analyze Apple's stock performance over the past year. Include: (1) total return vs S&P 500, (2) volatility compared to the market, (3) any notable events that caused big price moves. Present as bullet points with specific numbers."

Result: Focused, actionable, structured response

Prompt Engineering Principles

Master these five principles to dramatically improve your AI interactions:

1. **Be Specific**
   Instead of "analyze this," say "calculate the Sharpe ratio and compare it to the S&P 500 benchmark"

2. **Provide Context**
   Indicate your expertise level: "I'm a beginner" or "I'm preparing a board presentation for experienced investors"

3. **Give Examples**
   Show the desired format: "Format like this: Stock: AAPL, Return: 15%, Risk Level: High"

4. **Specify Format**
   Request structure: "Respond in bullet points" or "Give me executable Python code with comments"

5. **Iterate Continuously**
   If the first response isn't quite right, refine your prompt based on what you learned

Prompt Examples for Finance

Explaining Concepts

"Explain the Sharpe ratio to someone who has never taken a finance class. Use a simple real-world analogy first, then provide the formula with a worked example using actual stock data."

Analyzing Data

"I have Apple stock data showing: average daily return 0.05%, standard deviation 1.8%, total return 15%. Explain what these numbers mean in practical terms and whether this represents good performance compared to typical stocks."

Writing Code

"Write Python code using yfinance and pandas to: (1) download 1 year of AAPL and MSFT data, (2) calculate daily returns for both stocks, (3) compute their correlation coefficient. Include detailed comments explaining each step for beginners."

Where AI Fits in Finance

AI applications are transforming every corner of the financial services industry:

Trading	Pattern recognition, execution optimization, arbitrage detection
Risk Management	Predicting defaults, stress testing scenarios, anomaly detection
Research	Analyzing earnings calls, reading SEC filings, sentiment analysis
Customer Service	Chatbots, personalized advice, robo-advisors
Fraud Detection	Identifying suspicious transactions in real-time
Portfolio Management	Asset allocation, rebalancing, factor investing strategies

The AI Revolution in Finance

What's Changed with LLMs

- Read and summarize thousands of documents instantly
- Write first drafts of research reports and presentations
- Explain complex financial concepts to clients at any level
- Generate and debug code for sophisticated analysis
- Answer questions about regulations and compliance instantly

What AI Still Can't Do Well

- Predict market movements reliably (markets are inherently unpredictable)
- Replace human judgment entirely in high-stakes decisions
- Understand truly novel situations without historical precedent
- Take legal or ethical responsibility for financial decisions

Today's Lab Preview

Get ready for hands-on experience with real financial data and AI tools!

Pull Real Stock Data

Use yfinance to download historical prices for companies you're interested in

Calculate & Visualize

Compute returns, create charts, and identify trends in stock performance

Practice Prompts

Write effective prompts to analyze financial data using AI assistants

Generate Code

Ask AI to generate Python code, then run and modify it for your analysis

Reflect & Learn

Evaluate AI's capabilities and limitations in financial analysis

No prior Python experience needed—just run the cells and fill in your prompts!

Key Takeaways

Return

Percentage gain or loss on an investment over a specified period

Risk

Volatility—how much returns fluctuate, measured by standard deviation

Sharpe Ratio

Return per unit of risk—higher values indicate better risk-adjusted performance

Moving Average

Smoothed price trend revealing underlying momentum, used for trading signals

yfinance

Free Python library providing easy access to download real stock market data

Prompt Engineering

Writing better, more specific instructions produces dramatically better AI outputs

For Next Week

Required Readings

- Hilpisch, Chapter 4: Data-Driven Finance (pp. 81-100)
- Fama & French (1993), "Common Risk Factors"—read introduction and conclusion sections

Come Ready to Discuss

- How has big data fundamentally changed the way finance operates?
- What is the "efficient market hypothesis" and does it still hold?
- Can AI give investors an edge, or do markets adapt too quickly?

Next week's lab: Deep dive into advanced financial data analysis techniques

Questions?

Office Hours

Wednesdays 6-7pm (right before class)

Drop by for any questions or extra help

Email Contact

hersh@chapman.edu

Response within 24 hours

Course Materials

All slides, code, and resources available on Canvas

Check regularly for updates

Let's head to the lab!